

1 MES Development Suite

The MES Development Suite provides functions to customize the HYDRA Client MES Operation Center according to your requirements. The sections in the following provide general background information that you require for customizations and other extensions.

1.1 Activating the MES Development Suite

To activate or deactivate the MES Development Suite, use the main menu, menu item [Extras](#) → [MES Development Suite](#).

You can only activate the MES Development Suite (or the menu item), if the required function authorization and licenses are available. The licenses must be installed in the relevant system.

To activate the MES Development Suite, you must assign the function authorization "mds" to the user.

The following products must be purchased to activate the MES Development Suite.



For historical reasons, the **products** of the MES Development Suite are different to the **licenses**, which must actually be available in the system.

Product (price list)		License system) (in the
MDS-BAS	MES Development Business Applications & Services	MDS-BAP
MDS-RPD	MES Development Suite Report Design	MDS-RPB

One of the above licenses is enough to activate the menu item, but only with MDS-BAS, all available functions are available. With the product MDS-RPB, only the functions are available after activation that are required for the report design.

1.2 Applications on the MOC

The MOC provides many different functions. The functions are made available via applications. Applications can offer very different functions, but their structure is always the same. This is true for complex evaluation applications like the *Workplace overview* and for a simple editing dialogs like the one to edit *Units*.

These are the basic elements of an application:

- Toolbar with buttons to call functions
- Selection area or selection panel to parameterize data queries
- Area for detail applications where one or any number of detail application(s) may be presented.
- Data sources or DataControllers that provide data for the detail applications, i.e. that call (web) services on the server supplying the relevant data.

From a technical point of view, editing dialogs for creating, editing and copying of data records are also applications. But editing dialogs normally do not provide detail applications. They only have a single (selection) area to enter parameters that are used to call the relevant editing function (i.e. the relevant web service).

Note: Editing applications are normally generated using the application generator (included in the product "MES Development Suite – Business Applications").

1.3 Meaning of customization

You can use the MOC to create new applications and change existing functions via customization. You do not control the available functionality via programming, in particular with applications, but you edit customization files (.config), which are usually changed via specifically developed customization dialogs. These dialogs are integrated into the software and are enabled when you activate the MES Development Suite.

A separate section below provides general background information on customization settings and their distribution to the clients. Some special features of customization settings are described here, for example of applications or menu items.

The available authorizations of the user or the available licenses specify the changes that can be made to the customization settings. But in general, a normal user can also change these files and save a table layout that has been changed according to their own requirements, for example.

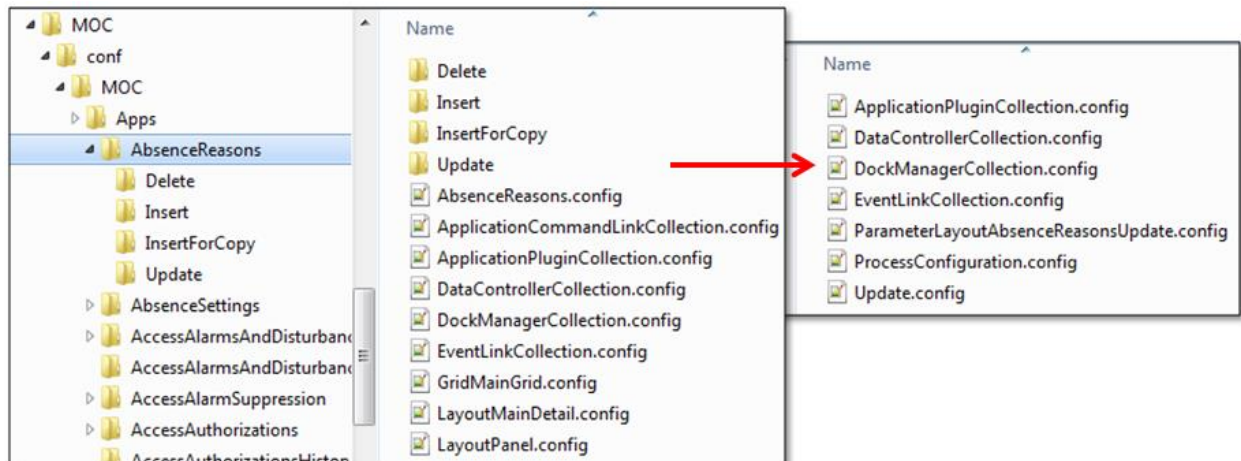
Customization files for applications

Some (main) applications are used to display data and other applications are used to edit data. For each main application, the directory %scope%\conf\Moc\Apps contains a directory with an unambiguous (English) name of the application. Editing applications are always assigned to a main application and are stored in a sub folder of this main application.

Example: The application *Absence reasons* is filed as a customization file in the "application" folder in the sub folder with the name of the related application ID, in this case *AbsenceReasons*. The editing dialogs required to edit absence reasons are stored in the sub folders "delete", "insert" and "update".

Main applications

The application directory contains all customization files that you require to customize an application. The specific files and the number of files are different for each application. This largely depends on the number of detail applications (tables, charts etc.) included in an application.



Contents of an application directory including editing applications.

The list below provides an overview of the most important files:

- <Id of the application>.config → Customization data for the application (title, help file, ...)
- LayoutPanel.config → Customization of the selection panel
- DockManagerCollection.config → Customization of the layout of detail applications
- DataControllerCollection.config → Customization of the data sources of the application
- ApplicationPluginCollection.config → Customization of the detail applications
- EventLinkCollection.config → Customization of the relations between data sources and detail applications
- ApplicationCommandLinkCollection → Customization of the toolbar.

For each detail application of a main application, a separate customization file is available. You can usually identify this customization file via the file prefix. For example

- Grid*.config → detail application with table
- *Chart*.config → Detail application with chart
- Pivot*.config → Detail application with pivot
- Layout*.config → detail application for detail views

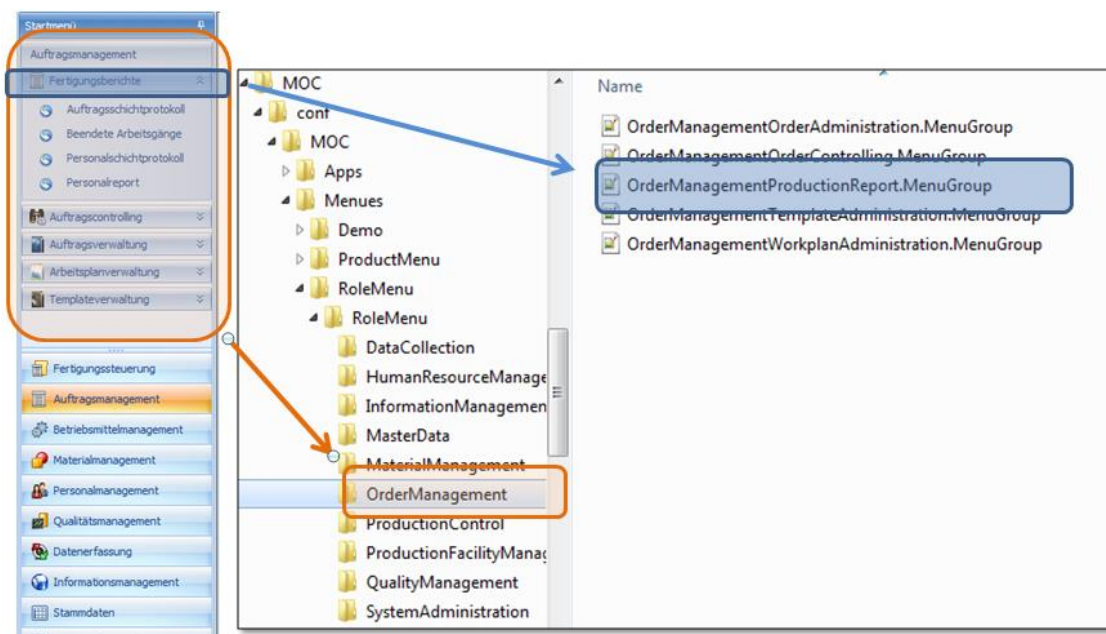
The application directory can include further files that are not listed here if special developments or plug-ins are available.

Editing applications

Editing applications are a special type of applications (see above). Their customization files are managed in sub folders of a main application. In addition to the files described above, the application directory of the main application contains an additional directory for each editing application including the customization files. Each editing application includes an additional file `ProcessConfiguration.config` that includes information on the process of calling this editing application. And the main customization file also includes additional and specific information.

Customization files for menus

On the MOC, you can use the menu item [Extras](#) → [Menu editor](#) to create and manage any number of menus. Each menu includes (main) menu items including sub items storing the actual functions. The customization files of the different menus are stored in the `%scope%\conf\Moc\Menues` directory. Each main menu is managed in a separate sub folder. The sub menus are mapped by a separate file containing the functions.



Example of a menu structure: The folder of the menu “RoleMenu” includes sub folders, e.g. the “OrderManagement” folder that manages the structure of the “order management” menu. The sub menu “production reports” and its functions are managed in the file “OrderManagementProductionReport.MenuGroup”.